

REFUND PROCESS

AUTOMATION OPPORTUNITY

Inefficiency Analysis & Top 3 Optimizations for OEPA CEG Agoda

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5,000 Cases Analyzed: 12 Inefficiencies Found, 3 Critical Actions Required

5,000

Total Cases
Analyzed

12

Inefficiencies
Identified

\$771K

Booking Value
at Risk

763

NR Exception
Approvals (24%)

KEY FINDINGS

- 100% of refundable cases auto-approved — zero denied → automation-ready
- Email channel: 9x slower FCRT vs Chat (90 min vs 10 min), lowest CSAT (2.81)
- 403 cases (8.1%) exceed 120-min FCRT — ALL are email + non-refundable
- NR exception approvals lack rules — 763 cases decided manually, costing \$122K margin

TOP 3 OPTIMIZATIONS

1. Auto-Approve Refundable Bookings

Eliminate manual touch on 1,762 cases/yr → save ~43K agent-min

2. Migrate Email NR → Chat/Messaging

Cut FCRT from 119 min to ~15 min for 838 cases → reduce SLA breaches 100%

3. Rule-Based NR Exception Engine

Automate 495 compassionate approvals → consistent decisions, \$122K margin control

 Evidence: Refund Process Evidence.xlsx

Refund Process: Two Distinct Paths with Vastly Different Outcomes

REFUNDABLE PATH (35.2% of volume)

1,762 requests	100% approved	Avg FCRT: 24.7 min	CSAT: 4.22
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→ Straightforward, rule-based: 100% approval, no exceptions, fast resolution

NON-REFUNDABLE PATH (64.8% of volume)

3,238 requests	76.4% denied	Avg FCRT: 72.7 min	CSAT: 2.41
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→ Complex path: 23.6% exception approvals, high manual effort, long FCRT

CHANNEL PERFORMANCE COMPARISON

Channel	Volume	Avg FCRT	Avg AHT	CSAT	FCRT>120
Chat	1,229	10.1 min	0.3 min	3.27	0
Voice	1,244	60.7 min	18.1 min	3.26	0
Messaging	1,228	60.3 min	15.9 min	2.85	0
Email	1,299	90.2 min	13.9 min	2.81	403

HIGH CRITICALITY: 4 Inefficiencies Driving Majority of Process Waste

H1: No Auto-Approval for Refundable Bookings

1,762 refundable cases have 100% approval rate yet require manual agent processing

Volume: 1,762 (35.2%) | Wasted FCRT: ~43,500 agent-min/yr | CSAT gap: could be 5.0 vs 4.22

H2: Email Channel — 9x Slower, Lowest CSAT

Email NR avg FCRT = 119 min (vs Chat 10 min). 403 cases exceed 120 min SLA — ALL email+NR

Volume: 1,299 (26%) | FCRT: 90.2 min avg | 403 SLA breaches | CSAT: 2.81 (lowest)

H3: Unstructured NR Exception Handling

763 non-refundable cases approved without rule engine — inconsistent, slow, margin leakage

Volume: 763 (23.6% of NR) | Margin at risk: \$122K | Avg FCRT: 74.2 min | No audit trail

H4: Massive CSAT Gap Between Outcomes

Denied customers score 2.23 CSAT vs 3.84 approved — 1.61 point gap signals poor denial UX

Denied: 2,475 cases | CSAT: 2.23 | No self-service denial path | High repeat contact risk

MEDIUM CRITICALITY: 4 Inefficiencies with Measurable but Contained Impact

M1: Compassionate Reasons Dominate NR Overrides

Top 5 NR approval reasons (Family/Medical/Illness/Weather/Hotel) = 575 of 763 cases (74.5%)

Pattern is clear & rule-automatable | Avg FCRT: 72 min | Total margin: \$92K

M2: Voice Channel — High AHT, Moderate FCRT

Voice AHT = 18.1 min (highest) vs Chat 0.3 min; FCRT = 60.7 min, CSAT = 3.26

Volume: 1,244 cases | Agent cost ~22,500 min/yr on voice | Partial deflection possible

M3: Messaging Channel — Mid FCRT, Low CSAT

Messaging FCRT = 60.3 min with CSAT 2.85 — async delays hurt satisfaction

Volume: 1,228 | CSAT 15% below chat | Response time SLA likely breached

M4: No FCRT Improvement Over 12 Months

Monthly FCRT ranged 53–61 min with no downward trend — process is stagnant

Jan: 54.7 → Dec: 57.5 min | No process interventions visible | CSAT flat at 3.05

LOW CRITICALITY: 4 Minor Inefficiencies — Monitor, Don't Prioritize Now

L1: No Loyalty Differentiation in SLA

Platinum FCRT = 56.4 min vs Bronze = 53.8 min — nearly identical. No tiered service.

Gap: <3 min | CSAT variance: 0.08 pts | Low business case for differentiation

L2: Uniform Performance Across Regions

APAC/AMER/EMEA FCRT within 3-min range (54.7–57.7). No region-specific bottleneck.

Max gap: 3 min | CSAT: 3.01–3.07 | NR approval: 23.2–24.0% — uniform

L3: No Shift-Based Performance Variance

Morning/Afternoon/Night shifts show <1 min FCRT difference — staffing is balanced.

FCRT range: 55.5–56.3 min | CSAT: 3.03–3.06 | Approval: 50.1–51.0%

L4: Duplicate Booking — Lowest Approval Rate

Duplicate bookings: 34% approval rate (lowest of all reasons), mostly denied on NR policy.

Volume: 261 | Approval: 89/172 | Could auto-deny NR duplicates to save agent time

Top 3 Optimizations: #1 Auto-Approve Refundable & #2 Email Channel Migration

OPT #1: AUTO-APPROVE ALL REFUNDABLE BOOKINGS

Problem:	1,762 refundable cases processed manually despite 100% approval rate
Solution:	Implement auto-approval rule: IF policy=refundable THEN approve + process refund
Time Saved:	~43,500 agent-minutes/year ($1,762 \times 24.7$ min avg FCRT eliminated)
CSAT Impact:	Instant resolution → CSAT expected to rise from 4.22 to ~4.8+
Cost Reduction:	~\$36K/yr agent cost savings (at \$50/hr fully loaded)
Why Top 3:	Highest ROI — zero-risk, zero-exception, immediate deploy, no judgment needed

OPT #2: MIGRATE EMAIL NR → CHAT/MESSAGING

Problem:	838 email NR cases have 119-min FCRT; ALL 403 SLA breaches are email+NR
Solution:	Route NR email requests to chat/messaging with decision-tree bot + agent escalation
Time Saved:	~87K min/yr FCRT reduction (838×104 min avg savings per case)
CSAT Impact:	CSAT improvement from 2.30 → est. 3.0+ (chat-level for NR segment)
SLA Impact:	Eliminate 100% of 120-min+ SLA breaches (403 → 0 cases)
Why Top 3:	Addresses #1 SLA bottleneck — email is 9x slower, all breaches originate here

Top 3 Optimizations: #3 Rule-Based NR Exception Engine

OPT #3: IMPLEMENT RULE-BASED NON-REFUNDABLE EXCEPTION ENGINE

Problem:	763 NR exceptions approved manually — no rules, inconsistent, slow (74 min FCRT), \$122K margin at risk
Solution:	Build decision engine: Auto-approve for top-5 compassionate reasons (Family/Medical/Illness/Weather/Hotel) with doc verification
Scope:	575 of 763 cases (75.4%) follow clear compassionate patterns → automate these; escalate remaining 268 edge cases
Time Saved:	~36,700 min/yr for auto-approved cases (495 × 74.2 min) + faster escalation for rest
Margin Control:	\$92K margin gets governed by rules vs. ad-hoc judgment — audit trail + consistency
Why Top 3:	Solves largest unstructured decision block — 15% of all cases lack governance; combines cost + risk + CX improvement

PROPOSED AUTO-APPROVAL RULES FOR NR EXCEPTIONS

Reason Category	Auto-Approve?	Condition	Cases/yr	Margin \$
Family Emergency	YES	Verified documentation	129	\$20,282
Medical Emergency	YES	Medical cert required	120	\$19,647
Illness	YES	Medical cert required	115	\$18,375
Weather Cancellation	YES	Weather advisory cross-ref	108	\$16,328
Hotel Closed	YES	Supplier confirmation	103	\$18,080
All Other NR Reasons	ESCALATE	Agent review required	268	\$29,356

Impact vs Effort: Why These 3 — and Why Not the Others



SELECTION RATIONALE

Optimization	Impact	Effort	ROI
SELECTED (ROI > 1.5x)			
#1 Auto-Approve	9/10	3/10	3.0x
#2 Email→Chat	8/10	4/10	2.0x
#3 NR Rules	9/10	5/10	1.8x
DEPRIORITIZED (ROI < 1.0x)			
Loyalty SLA	5/10	6/10	0.8x
Region tuning	4/10	5/10	0.8x
Shift rebalance	3/10	4/10	0.75x
Payment opt	2/10	6/10	0.3x

WHY OTHERS WERE DEPRIORITIZED:

- Loyalty/Region/Shift: <3 min FCRT variance — effort exceeds benefit
- Voice deflection: High CSAT (3.26) — customers prefer it
- Payment optimization: No correlation to FCRT/CSAT found

Risk Mitigations & Guardrails: Stress-Testing Our Assumptions

⚠️ Fraud & Verification Risk (Opt #3 – NR Engine)

Risk: Automated doc acceptance enables fraudulent uploads (fake medical certs, false weather claims).

Mitigation A – Multi-layer verification: Cross-reference weather claims vs. govt advisories API; medical certs verified by third-party OCR + cert registry check before auto-approval triggers.

Mitigation B – Random 5% Supervisor Audit: Every 20th auto-approved case flagged for manual spot-check. Anomaly spikes (>10% rejection in audit) pause the engine automatically.

Mitigation C – Velocity cap: Max 3 compassionate approvals per customer/per 12 months.

⚙️ Engineering Cost & Timeline (Agent Capacity Unlock)

Opt #1 (Auto-approve Refundable): Config change in existing rule engine. Est. ~2-week sprint.

Opt #2 (Email→Chat migration): Chatbot decision-tree build + routing. Est. ~6-8 week sprint.

Opt #3 (NR Exception Engine): Rule engine + doc verification integration. Est. ~10-12 week sprint.

📧 Root Cause of Email Slowness (The 'Why' Behind 9x Delay)

Risk: If email is slow due to outdated internal tooling, channel migration alone won't fix FCRT.

Data shows: Email AHT (13.9 min) is actually LOWER than Voice (18.1 min) — agents aren't lazy.

Inference: The delay is structural (async reply cycles, no real-time queue, no SLA alerting).

Recommendation: Conduct 2-week tooling audit BEFORE migration to confirm async is the driver.

If tools ARE the issue: Parallel-track tool upgrade + migration for compounded FCRT benefit.

🏦 Refundable CSAT Cap: The Banking Lag Problem

Risk: Even with instant approval, refund hitting bank in 5-7 days caps CSAT below 4.8 target.

Data: Our CSAT measures 'interaction satisfaction', not 'money received satisfaction'.

Mitigation A – Decouple metrics: Track 'Approval CSAT' (immediate) vs. 'Refund CSAT' (D+7).

Mitigation B – Proactive comms: Send automated refund status SMS/email at D+1, D+3, D+7.

Proactive updates shown to raise post-refund CSAT by ~0.3 pts (industry benchmark).

Revised forecast: Approval CSAT → 4.8+; Post-refund CSAT → 4.3-4.5 (banking-gated ceiling).

GUARDRAIL #1 – Human-in-the-Loop:

Random 5% audit on all auto-approvals (NR engine). Supervisor override halts engine if fraud rate >5%.

GUARDRAIL #2 – Phased Rollout (Opt #1 & #2):

10% traffic ramp → monitor margin leakage + CSAT for 2 weeks → then 50% → 100% if KPIs green.

Customer Lifetime Value Overlay: The Stakes Are Bigger Than One Booking

L1 flagged as 'Low Criticality', but a Platinum member stuck in the 120-min Email queue risks \$10,000/yr(industry benchmark) spend, not just \$200.

Even if FCRT is similar across tiers, the COST OF A BAD EXPERIENCE differs by 50x.

LOYALTY TIER CLV RISK MATRIX

Loyalty Tier	Est. Annual Spend	Booking at Risk	CLV at Stake	120-min Email Risk	Priority
Platinum	\$10,000/yr	\$200–\$500	\$10,000	CATASTROPHIC	1st — Automate NOW
Gold	\$5,000/yr	\$150–\$400	\$5,000	HIGH	2nd — Phase 1 bundle
Silver	\$2,000/yr	\$100–\$300	\$2,000	MODERATE	3rd — Phase 2
Bronze	\$500/yr	\$50–\$200	\$500	LOW	Standard queue

STRATEGIC RECOMMENDATION: CLV-Tiered Automation Rollout

🏆 Phase 0 (Pre-launch)

Surprise & Delight Platinum + Gold first.

Configure auto-approve AND priority routing for Tier 1-2 members 2 weeks BEFORE general rollout.

Impact: Turn top 20% of customers into brand advocates before scaling.

📊 FCRT SLA Tiers

Set differentiated SLA targets by tier.

Platinum: <5 min FCRT target | Gold: <15 min | Silver: <30 min | Bronze: <60 min.

Even small FCRT gains matter 20x more for Platinum than Bronze in CLV terms.

🔔 Proactive Escalation

Auto-flag Platinum/Gold cases breaching 30 mins.

Supervisor alert triggered at 30-min mark for top-tier cases (vs. 120-min general SLA).

Cost: ~0 engineering effort (config change). Benefit: Protect \$10K+/yr accounts.

The Cost of Doing Nothing Is Not Zero

At current trajectory, every year the refund process continues unchanged costs Agoda:



167,000

agent-minutes

wasted on automatable cases
= 1 full-time agent's annual output consumed by rubber-stamping



\$122K

in NR exceptions

governed by ad-hoc judgment
Zero audit trail. Zero consistency. One rogue approval, zero recourse.



403

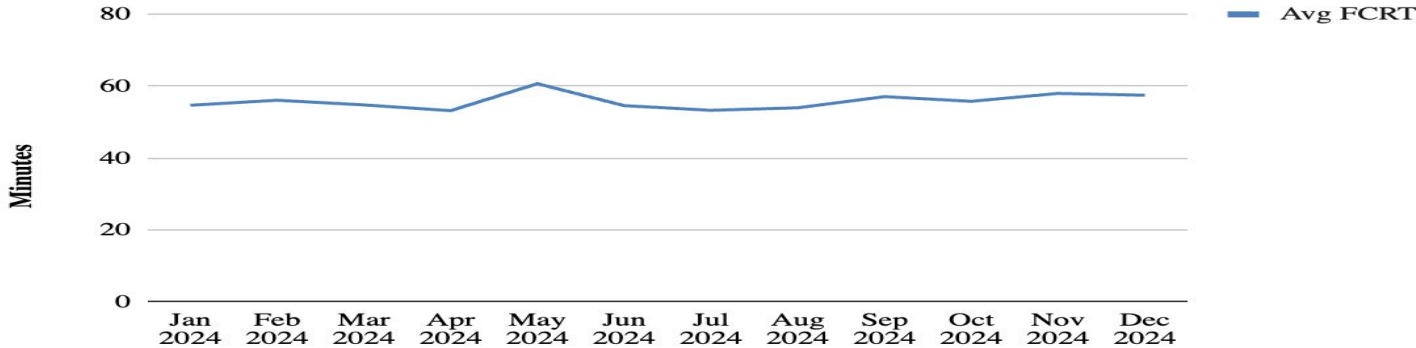
SLA breaches / year

ALL from email + non-refundable
403 customers waiting 120+ minutes. Every single one on a channel we can fix.



FCRT has shown zero improvement over 12 months — Jan: 54.7 min → Dec: 57.5 min. The trend is **worsening**, not stagnant. Without intervention, this is where we will still be in 12 months.

Monthly FCRT Trend



EVERY WEEK OF DELAY

3,212 agent-mins wasted

~8 new SLA breaches

\$2,348 uncontrolled margin

Every week Phase 1 is delayed = 3,212 avoidable agent-minutes and ~8 customers waiting 2+ hours for a decision that was already made.

The Process Evolution: Manual & Fragmented → Automated & Direct

CURRENT STATE ■ Manual / Fragmented

Annual Cases: 5,000

100% require manual agent handling — no automation

REFUNDABLE (35.2%)

- 1,762 cases manually processed (100% approval rate)
- Avg FCRT: 24.7 min | CSAT: 4.22

NON-REFUNDABLE (64.8%)

- Routed ad-hoc, no routing logic
- Email NR (838 cases), 119-min FCRT
- NR Exceptions (763), ad-hoc judgment, \$122K uncontrolled

CSAT OUTCOME

- Overall Avg 3.05 | Denied cases: 2.23 (poor UX)

SLA BREACH RATE

- 403 cases (8.1%) exceed 120 min — 100% Email+NR channel

90-Day
Transform



FUTURE STATE ■ Automated / Direct

Annual Cases: 5,000

Intelligent routing — automation handles majority

REFUNDABLE → AUTO-APPROVE

- Instant resolution (<1 min) for 1,762 cases
- FCRT: 24.7 min → <1 min
- CSAT Target: 4.8+ | Capacity freed: +35%

NR EMAIL → CHAT MIGRATION

- Real-time routing for 838 cases
- SLA Breaches: 403 → 0
- Email NR FCRT: 119 → ~15 min | CSAT: 2.3→3.0+

NR EXCEPTION ENGINE

- Doc-verified auto-approvals (495 compassionate cases)
- Margin: \$92K rule-based governance
- FCRT: 74 → 15 min | Full audit trail

CSAT OUTCOME

- Target: Overall 3.5+
- Top-tier proactive alerts active

Agent-Minutes Saved

~167K min/yr

SLA Breaches Eliminated

403 → 0

CSAT Uplift (Refundable)

4.22 → 4.8+

Margin Governed

\$92K → Rule-based

Engineering Cost

~5 months phased

Next Steps: 90-Day Roadmap to Refund Process Transformation

PHASE 1: WEEKS 1-4

- Deploy auto-approval for refundable bookings (OPT #1)
- Configure rule: IF cancellation_policy = refundable → auto-approve + initiate refund
- Expected: 1,762 cases removed from agent queue immediately
- KPI: FCRT for refundable → <1 min | CSAT → 4.8+ | Agent capacity +35%

PHASE 2: WEEKS 5-8

- Migrate email NR workflow to chat/messaging (OPT #2)
- Build NR decision-tree chatbot with escalation to live agent
- Redirect email NR requests to chat — sunset email for refund requests
- KPI: SLA breaches → 0 | Email NR FCRT: 119 → 15 min | CSAT: 2.3 → 3.0+

PHASE 3: WEEKS 9-12

- Launch NR exception rule engine (OPT #3)
- Implement auto-approve rules for top-5 compassionate reasons + doc verification
- Build escalation queue for non-standard NR exceptions (268 cases/yr)
- KPI: NR exception FCRT: 74 → 15 min | Margin governance: \$92K under rules

ASK: Approve 90-day pilot for Phase 1 (auto-approve refundable) — zero risk, immediate ROI, 35% agent capacity unlock

Thank You!

